Post partum Hemorrhage

Topic 7

Identification of PPH

- Loss of 500 ml or more of blood during delivery and up to six weeks after delivery (may be less in anemia)
- or
- Blood loss sufficient to cause signs and symptoms of hypovolemia
- or
- Woman soaks 1 pad or cloth in <5 min

CAUSES OF PPH

(Postpartum Hemorrhage)

PPH is excessive bleeding after childbirth.
Understanding the causes helps in timely prevention and management.

PRIMARY / IMMEDIATE PPH

Occurring during delivery till
24 hours postpartum

Tone - Atonic PPH
Most common cause (80-90%)

Tears or trauma
Lacerations of the cervix,
vagina, perineum, or uterus

Tissue - Retained or incomplete placenta, membranes
Placenta or membranes not
delivered completely

Thromboembolic - Coagulopathy
Bleeding due to clotting disorders

SECONDARY / DELAYED PPH

From 24 hours postpartum till
42 days or 6 weeks

Infection in the uterus
Postpartum endometritis
or uterine infection

Retained placental fragments
Placental tissue left behind
causing bleeding

Early recognition and management of the cause can save lives.

PRINCIPLES OF MANAGEMENT OF PPH

1 CALL FOR ADDITIONAL SUPPORT

Activate help early – alert senior staff and prepare for possible escalation.

2 MANAGE SHOCK

Assess ABCs, give oxygen, establish IV access, start fluids and monitor vitals.

3 CONTINUE UTERINE MASSAGE

Massage the uterus firmly and continuously to stimulate contraction.

4 TREAT SPECIFIC CAUSES OF PPH

- **Try medical (uterotonics) and conservative management** (such as bimanual compression, aortic compression, balloon tamponade) before conducting surgical procedures.

Uterotonics | Bimanual Compression | Aortic Compression | Balloon Tamponade

5 STABILIZE THE PATIENT BEFORE REFERRAL

Patient's condition should be stabilized before any referrals are done.

6 IF REFERRAL IS NEEDED

Continue fluids, uterotonics and temporizing measures if needed to control the bleeding.

REMEMBER: The interval from the onset of PPH to death can be as little as two hours, unless appropriate life-saving steps are taken immediately.

IDENTIFICATION OF SHOCK

Shock is a life-threatening condition caused by inadequate oxygen and blood flow to the body’s organs.

SIGNS OF SHOCK

1 TACHYCARDIA
Fast, thin, thready pulse, >110/minute

2 TACHYPNOEA
Fast respiratory rate

3 HYPOTENSION
Fall in systolic BP, <90 mm of Hg

4 HYPOTHERMIA
Skin cold and clammy

5 ALTERED SENSORIUM
Drowsy, semi conscious or unconscious

- Drowsy
- Semi conscious
- Unconscious

ACT EARLY. SAVE LIVES.
Early recognition and prompt management of shock can prevent organ failure and death.

A Pictorial Reference Guide to Aid Visual Estimation of Blood Loss at Obstetric Haemorrhage: Accurate Visual Assessment is Associated with Fewer Blood Transfusions

Dr Patrick Bose, Dr Fiona Regan, Miss Sara-Paterson Brown

Soiled Sanitary Towel
30ml

Soaked Sanitary Towel
100ml

Small Soaked Swab 10x10cm
60ml

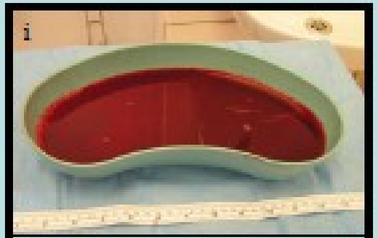
Incontinence Pad
250ml

Large Soaked Swab 45x45cm
350ml*

100cm Diameter Floor Spill
1500ml*

PPH on Bed only
1000ml

PPH Spilling to Floor
2000ml



Full Kidney Dish
500ml

COLOUR	GAUGE (G)	FLOW RATE (ml/min)	COMMON USE
ORANGE	14 G (2.1mm)	240 min	Major trauma, rapid fluid resuscitation
GREY	16 G (1.7mm)	180 min	Emergency surgery, blood transfusion
GREEN	18 G (1.3mm)	90 min	IV fluids, blood transfusion
PINK	20 G (1.1mm)	60 min	Medications, maintenance fluids
BLUE	22 G (0.9mm)	36 min	Elderly, pediatric patients
YELLOW	24 G (0.7mm)	20 min	Neonates, very fragile veins
PURPLE	26 G (0.6mm)	13 min	Neonates

Classification of hypovolemic shock

Signs	Class I	Class II	Class III	Class IV
Blood loss (mL)	500-1000 15%	1200-1500 20-25%	1800-2100 30-35%	> 2400 > 40%
Pulse/min	Normal	100	120	140
Systolic BP (mm Hg)	Normal	Normal	70-80	60
Tissue Perfusion	No symptoms or signs		Pallor, rapid respiration, restlessness, oliguria, cold skin	Collapse, anuria, Rapid RR (air hunger), cold skin
Mental status	Normal		Agitated response	Confused, agitated, aggressive
The blood loss is replaced by IV fluids. IV fluids should be given when losses amount to 700mls Initially give 1 lt in 20 min and decide further fluid requirement based on the vital parameters				

Blood Volume 65 ml/Kg

Assessment of Blood Loss

- Recognize PPH by correctly assessing the blood loss after childbirth.
- Until the woman loses up to 1000 ml of blood her pulse rate, BP, general condition can appear as normal
- Pulse 100 /min and BP normal → blood loss can be 1200 ml or more
- Pulse 120/min , systolic BP is 80 mm Hg or less, woman is pale, cold, restless and agitated : Blood loss is 1800 – 2000 ml
- Woman with low BMI Blood volume is less
- Severe PE/Eclampsia
- Moderate/severe anaemia

USE OF CLIBRATED DRAPE FOR BLOOD LOSS ESTIMATION

Active Management of Third Stage of Labour (AMTSL) for Prevention of Postpartum Haemorrhage (PPH)

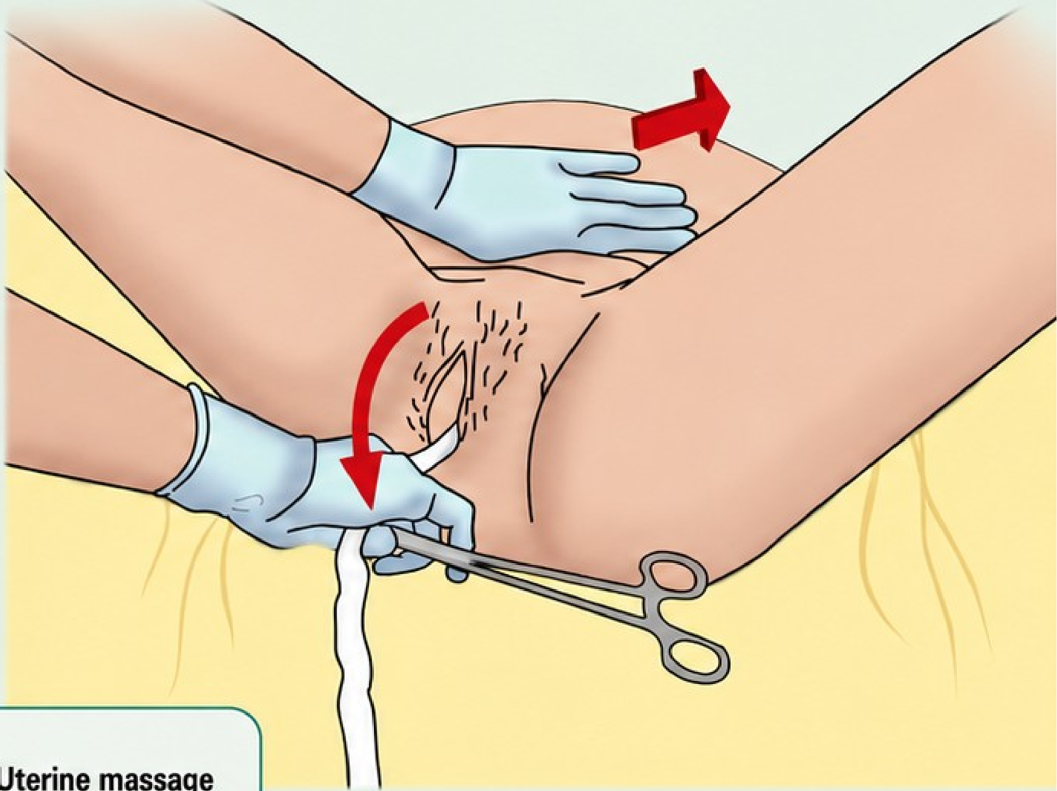
Prevention of PPH

- Mandatory for all deliveries (vaginal and abdominal)
- Exclude presence of another baby after delivery of first baby

Step 1 Inj. Oxytocin 10 units IM immediately after birth

Step 2
- Controlled cord traction once uterus is contracted and cord is cut
- Apply cord traction (pull) downwards and give counter-traction with other hand by pushing uterus up towards umbilicus

Step 3 Uterine massage to keep uterus contracted



KEY TO PREVENTION OF PPH

1 Uterine contraction (Oxytocin) → 2 Controlled cord traction → 3 Uterine massage

INITIAL MANAGEMENT OF PPH

1 SHOUT FOR HELP

Mobilize all available health personnel.

2 EVALUATE VITAL SIGNS

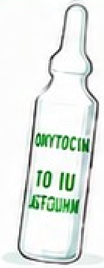
Pulse, BP, respiration and temperature.

3 ESTABLISH TWO IV LINES

Use wide bore cannula (16–18 gauge), draw blood for blood grouping and cross matching; catheterize the bladder.

4 START RAPID INFUSION

Using cannula number 16 with Normal Saline/ Ringer Lactate, 1L in 15–20 mins.



5 GIVE Inj. OXYTOCIN

10 IU I/M

(if not given during AMTSL)

6 START Inj. OXYTOCIN

20 IU in 500 ml RL

@ 40–60 drops/ Min.

7 ADMINISTER Inj. TRANEXAMIC ACID (TXA)

1g IV

8 GIVE OXYGEN

@ 6–8 L per minute by mask.

9 MONITOR VITAL SIGNS AND BLOOD LOSS

Every 15 minutes.

10 MONITOR FLUID INTAKE AND URINARY OUTPUT

Use urinary catheter and urobag to monitor output.

11 CHECK IF PLACENTA HAS BEEN EXPELLED – MANAGE CAUSE SPECIFIC PPH

Ensure complete expulsion of placenta and membranes.

Identify and manage the cause of PPH.

Cause Specific Management: Atonic PPH

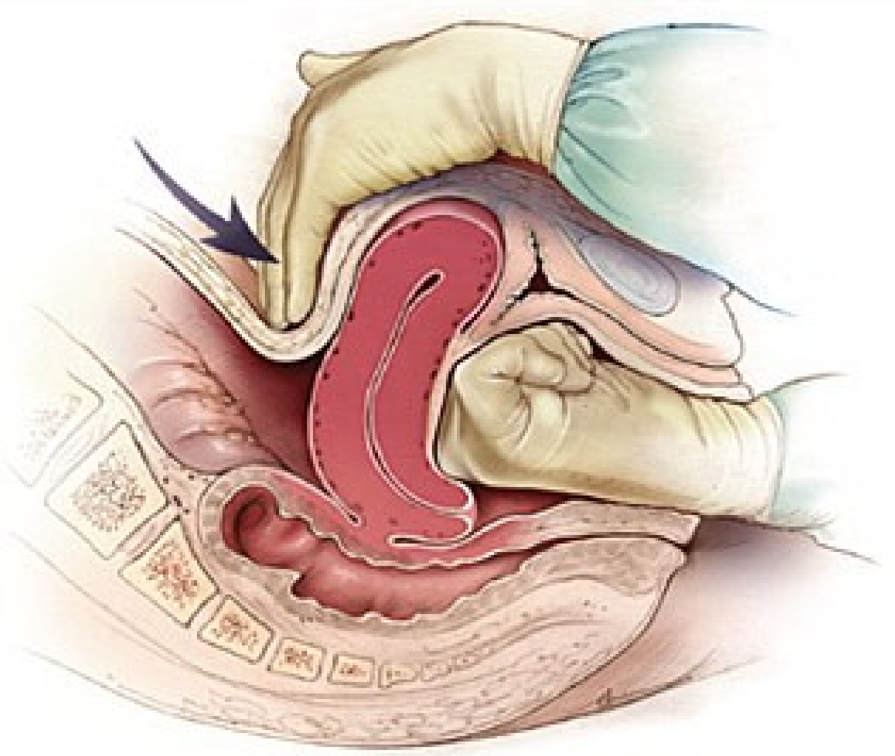
- Catheterize and keep in dwelling catheter in situ.
- Massage uterus to stimulate effective uterine contractions
- If still uterus is relaxed can give other uterotonics like prostaglandins (misoprostol or carboprost) or methyl ergometrine or combination of oxytocin and methyl ergometrine
- If uterus is still relaxed- examine placenta for completeness, if placenta complete and uterus still relaxed, perform bimanual uterine compression/aortic compression as a temporary measure, perform condom tamponade (at facilities where medical officers available)
- Arrange for transportation to FRU where facilities for blood transfusion and surgery available
- If bleeding is controlled by drugs- repeat uterine massage every 15 min for first 2 hrs, closely monitor vitals, continue oxytocin (total not exceeding 100 IU in 24 hrs)

DRUGS FOR PPH MANAGEMENT

DRUG	DOSE AND ROUTE	CONTINUE DOSE	MAX DOSE	PRECAUTION AND CONTRAINDICATION
Oxytocin	• 20 IU in 500ml RL • 40–60 drops • IV infusion	• IV infuse 20 IU in 500ml RL • 40–60 drops / mins • IV infusion	Not more than 3L of IV fluids containing Oxytocin	• Do not give IV as bolus & do not exceed 100 IU in 24 hours
Tranexamic acid (TXA)	• 1 g IV	• Single dose	Single dose (1gm)	• Thromboembolic disorders, Active thromboembolism
Ergometrine	• IM or IV (slowly) 0.2 mg	• Repeat 0.2 mg after 15 min. If required give 0.2 mg IM/IV slowly every 4 hrs	Five dose (Total 1.0mg)	• High BP, PE, Heart disease
15- Methyl prostaglandin F2-alpha	• IM 0.25 mg	• 0.25 mg every 15 min	8 doses (total 2mg)	• Asthma
Misoprostol PGE-1	• 800 micrograms PR/sublingual	• Single dose	Single dose	-

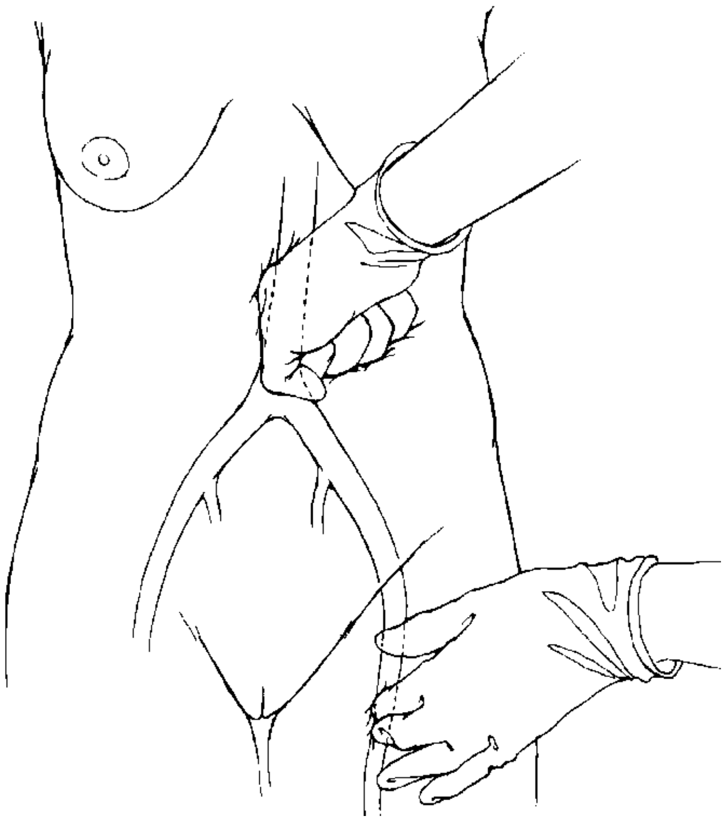
Bimanual Uterine Compression

- Empty urinary bladder with Foley's catheter
- Insert gloved hand in vagina, remove any visible clots from vagina
- Place fist in anterior vaginal fornix and press against anterior wall of uterus
- Place other hand on abdomen behind uterus, pressing against posterior wall of uterus
- Maintain compression until bleeding is controlled and uterus contracts



Compression of Abdominal Aorta

- Apply downward pressure with closed fist over abdominal aorta directly through abdominal wall
- With other hand, palpate femoral pulse to check adequacy of compression
  - Pulse palpable = inadequate
  - Pulse not palpable = adequate
- Maintain compression until bleeding is controlled



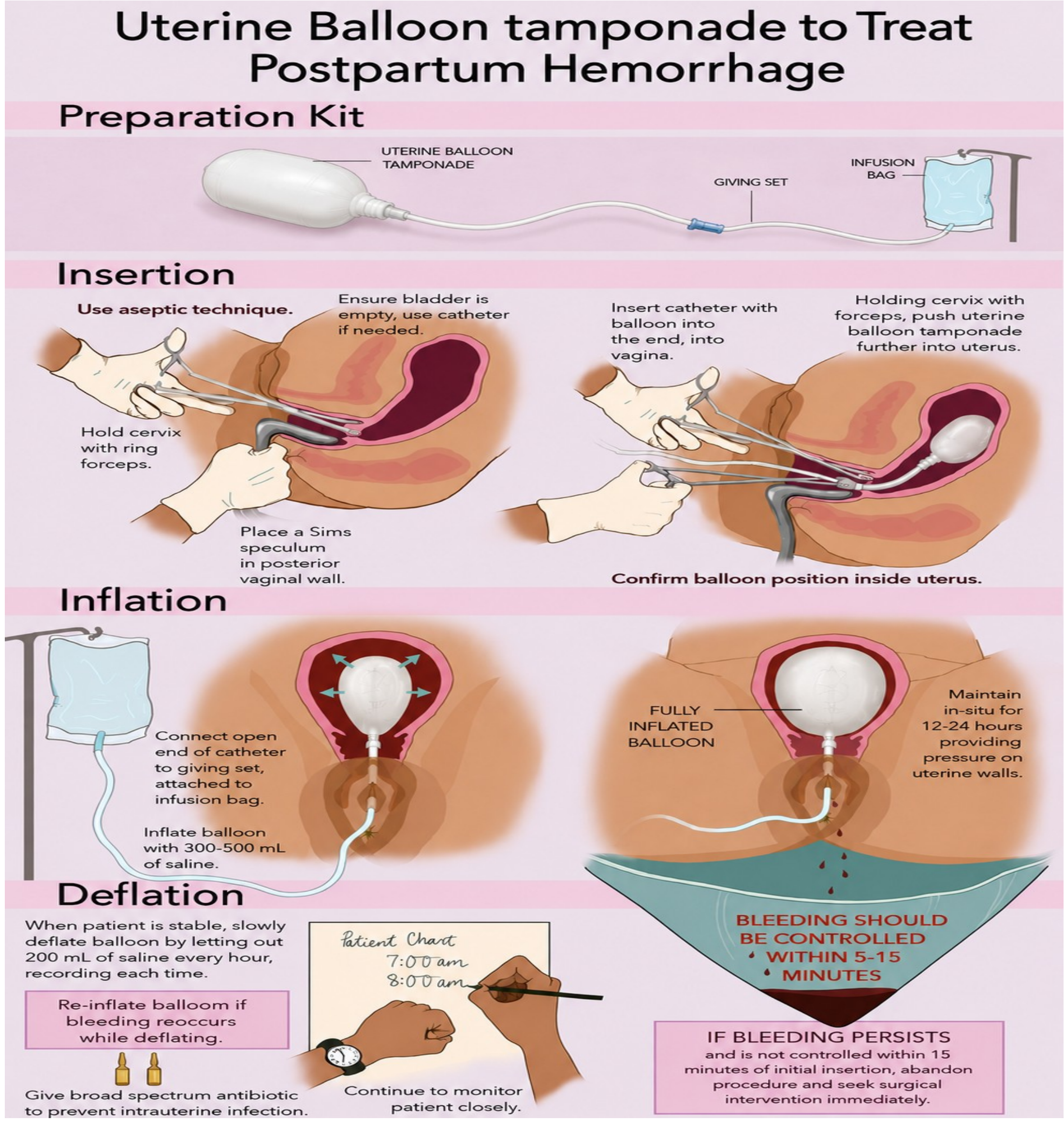
Condom Tamponade

Insertion:

- Ensure that the bladder is empty.
- Hold cervix with a ring forceps.
- Place a Sims speculum in posterior vaginal wall.
- Insert catheter with condom tied onto the end (tied using sterile suture), into the vagina.
- Holding cervix with forceps, push condom further into uterus.

Inflation:

- Connect open end of catheter to IV set attached to infusion bag & inflate with 300 to 500 ml saline.
- Clamp catheter after inflating.
- Maintain in-situ for 12 to 24 hours.
- Keep bladder empty by indwelling Foley's, put on woman on prophylactic antibiotics.
- Monitor the patient closely.



NON-PNEUMATIC ANTI-SHOCK GARMENT (NASG)

WHAT IS NASG?

An external compression device that reduces blood flow to the lower body, maintains blood pressure and reduces blood loss.

Place it under the woman; ensure top of the device is below the lowest rib.

APPLICATION STEPS (BY TWO PERSONS)

1. Start at the ankles; close segment 1 around both ankles tightly; tightness is felt by a snap.
2. Next segment 2 around calves; same tightness.
3. Next segment 3 around thighs leaving the knees free.
4. Next segment 4 around pelvis at level of pubic symphysis.
5. Next segment 5 containing the foam ball over the lower abdomen directly over umbilicus so that the foam ball presses over the uterus.
6. Next segment 6 over the segment 5.

NOTE Fold the segment 1 if women is short.

REMOVAL (Only under medical supervision)

1. Remove segment one and WAIT for 15 min; record vitals.
2. If BP does not fall by 20 mmHg or pulse increase by 20 beats/min.
3. Then move on to removal of segment 2 and wait 15 min; recheck vitals; then segments 3, 4, 5 & 6 waiting 15 min each time to check BP & Pulse.
4. If BP falls by 20 mm Hg or pulse increases by 20 beats stop removal.
5. Starting with the last segment opened rapidly reapply from top to bottom and look for source of bleeding.

Cause Specific Management: Tears or Trauma

Suspect tears in case of contracted uterus with PPH:

- Look perineum, cervix and vagina for any tears or laceration
- In case of 1st degree perineal tears apply pressure through perineal pads
- In case of 2nd, 3rd or 4th degree perineal tears or cervical tears. Cover tear with sterile pad, establish IV line , infuse fluids rapidly, raise foot end of stretcher, keep her warm during transportation and refer woman to higher center for suturing

Key Messages

- PPH is the most important causes of maternal deaths
- 70% chances of PPH can be prevented by doing AMTSL for every delivery
- Deaths from PPH can occur within 2 hours of its occurrence, so timely identification, management and/or referral is very important
- If the woman is in shock, manage it and stabilize her on priority then manage PPH according to the cause
- Major cause of PPH is due to atonic uterus which can be prevented by AMTSL and early initiation of breastfeeding
- Secondary PPH is mainly due to infection of uterus so apart from PPH management, antibiotics will be required in such cases if the woman has fever and foul smelling lochia